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Xingda Lyu

Education

Expected Jun 2027 **B.S. in Statistics, Data Science Track**, *University of Washington*, Seattle, WA.

Expected Jun 2027 **B.S. in Informatics, with Honors**, *University of Washington*, Seattle, WA.
Cumulative GPA: 3.9/4.0
Selected coursework: Statistical Machine Learning, Mathematical Statistics & Probability I–III, Generalized Linear Models, Statistical Computing, Data Structures and Algorithms, Databases and Data Modeling, Real Analysis.

Publications & Presentations

Manuscripts

Manuscript **Lyu, Xingda***; Kaur, Kirandeep*; Zhou, Zhiyu; Kleiman-Weiner, Max; and Shah, Chirag, 2026. *Modeling Evolving Minds: Toward Epistemic Human-AI Collaboration.* Manuscript in preparation.

*Equal contribution.

In Conference Proceedings

ICML 2026 Kaur, Kirandeep; **Lyu, Xingda**; and Shah, Chirag, 2026. *Position: Knowing Isn't Understanding: Re-grounding Generative Proactivity with Epistemic and Behavioral Insight.* In Proceedings of the International Conference on Machine Learning (ICML 2026). (*paper*)

CDS 2025 **Lyu, Xingda**; Lyu, Gongfu; Yan, Zitai; and Jiang, Yuxin, 2025. *P-RAG: Prompt-Enhanced Parametric RAG with LoRA and Selective CoT for Biomedical and Multi-Hop QA.* In Proceedings of the International Conference on Computing and Data Science (CONF-CDS 2025). (*paper*)

Workshop Papers

CVPR 2026 **Lyu, Xingda**; Luo, Xinye; Lu, Honglin; and Yang, Shiqi, 2026. *PROS: A Proactive Multimodal Refinement Agent for Scientific Posters.* In Proceedings of the Workshop on Human-Interactive Generation and Editing (HiGen), CVPR 2026.

Selected Presentations

CHIIR 2026 **Lyu, Xingda**, 2026. *Decision-Making Under Unknown Unknowns in Proactive Agentic Systems.* Lightning talk and poster presentation, ACM CHIIR Workshop on Human-Centered Proactive & Personalized Agents.

ARD 2026 Kaur, Kirandeep; Gupta, Vinayak; **Lyu, Xingda**; Gupta, Aditya; and Shah, Chirag, 2026. *Proactive Agents for Knowledge Gap Navigation.* Poster presentation, Amazon Research Day 2026, AI Co-Scientist. (*poster*)

Research Experience

University of Illinois Urbana-Champaign

Summer 2026 ***Research Intern, Data Mining Group***.
Conduct research on AI for science, focusing on retrieval-augmented generation for retrosynthesis. Build and evaluate LLM/retrieval workflows for chemical reasoning and scientific decision support.

Advisor : **Prof. Jiawei Han**, *Michael Aiken Chair Professor, Siebel School of Computing and Data Science*, UIUC (*Personal Web-page*)

University of Washington

Nov 2025 – Present **_Research Assistant, InfoSeeking Lab_**.
Study proactive AI agents for information seeking, knowledge-gap navigation, and decision-making under uncertainty. Design benchmarks and model-based evaluations to analyze how agentic systems act under incomplete context and unarticulated user needs.

Advisor : **Prof. Chirag Shah**, *Professor, Information School; Adjunct Professor, Paul G. Allen School of CSE*, University of Washington (*Personal Web-page*)

Apr 2026 – Present **_Research Collaborator, UW NLP Group_**.
Research calibration and personalization challenges in proactive LLM assistants. Contribute to problem formulation, prompt and example design, model-output analysis, qualitative error analysis, and writing.

Collaborator : **Stella Li**, *Ph.D. Student, Paul G. Allen School of CSE*, University of Washington (*Personal Web-page*)

Sep 2025 – Present **_Research Project Lead, PROS_**.
Lead a self-directed project on proactive multimodal refinement of scientific posters using editable PPTX representations. Build source-grounded LLM/VLM diagnosis and constraint-aware editing methods for layout, structure, and scientific communication repair.

Project : *Independent Research Project*, University of Washington

Carnegie Mellon University

Nov 2024 – May 2025 **_Research Project Lead, P-RAG_**.
Led a research project on retrieval-augmented biomedical and multi-hop question answering. Integrated retrieval, LoRA-based parametric adaptation, selective chain-of-thought prompting, and data augmentation into an experimental QA pipeline.

Advisor : **Prof. David P. Woodruff**, *Professor, Theory Group, Department of Computer Science*, Carnegie Mellon University (*Personal Web-page*)

Professional Experience

Jun 2025 – Aug 2025 **_Data Science Intern_**, *Amazon*, Seattle, WA.
Built automated data-quality pipelines over 300K+ transportation records using Python and SQL. Developed routing optimization and anomaly-detection analyses to identify operational inefficiencies, refund gaps, and underperforming carriers.

Jul 2024 – Sep 2024 **_Data Science Intern_**, *Baynovation*, San Jose, CA.
Built a refinance decision tool using real-time mortgage rate data. Developed an interactive mortgage-scenario dashboard and delinquency-risk models using GLM, Random Forest, and XGBoost.

Honors & Patents

Award **_iSchool Conference Award_**, University of Washington ($1,500), 2026.

Patent **CNIPA-granted utility patents** for automated veterinary pharmaceutical production systems, co-inventor, 2023.

Patent **Chinese invention patent** for a microbial agent for preventing wheat powdery mildew, co-inventor, 2022.

Skills

Languages Python, R, SQL, Java, JavaScript, HTML/CSS, LaTeX

ML / AI PyTorch, Hugging Face Transformers, scikit-learn, LoRA/PEFT, RAG, LLM fine-tuning, multi-modal LLM/VLM systems

Research Information retrieval, human-AI interaction, proactive agents, AI for science, benchmark design, prompt and example design

Evaluation LLM evaluation, ablation studies, statistical modeling, qualitative error analysis, experimental analysis

Tools Git/GitHub, Linux, Jupyter Notebook, Flask, Overleaf, PowerPoint/PPTX